

Viet Nguyen

Machine Learning Engineer · Data Scientist

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SUMMARY

Machine Learning Engineer and Data Scientist building end-to-end ML pipelines, RAG systems, LangChain conversational AI, and serverless generative AI on AWS. M.S. in CS (Artificial Intelligence), Georgia Tech; AWS Certified Cloud Practitioner.

SKILLS

Languages & Tools: Python, SQL, Git, GitHub, Jupyter Notebook

Machine Learning: scikit-learn, Random Forest, Decision Tree, Gradient Boosting, Feature Engineering, SMOTE, ML Pipelines, Hyperparameter Tuning, GridSearchCV, Classification, Regression

Generative AI & LLMs: LangChain, ChatBedrock, Amazon Bedrock, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM APIs, Conversational AI, Vector Databases, Titan Embeddings, OpenSearch Serverless

Data Science: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, A/B Testing

Cloud & AWS: AWS Lambda, Amazon API Gateway, Amazon S3, Amazon EMR, AWS EC2, CloudWatch, IAM, boto3, AWS Certified Cloud Practitioner

Model Deployment & APIs: FastAPI, Pydantic, REST API, Model Serving, joblib

EXPERIENCE

Marketing Data Manager · Volunteer

2025 – Present

Association of Talent Development – Orange County (ATD-OC) · Anaheim, CA

- Delivered data-driven digital marketing analysis using an AI-powered agentic workspace, surfacing actionable growth opportunities across member engagement channels
- Developed a 3-phase strategic roadmap targeting 15% membership growth and 40% engagement improvement; deployed interactive analytics dashboard on Vercel

Engineering & Robotics Teacher · VEX AI Robotics Coach

2016 – Present

Jeffrey Trail Middle School · Irvine, CA

- Manage a \$15,000–\$20,000 annual program budget; teach AI, ML, robotics, and CS to middle school students
- Coached teams to 4 consecutive World Championships and 5 consecutive State Championships

Startup Co-Founder & Operations Lead

2024 – 2026

TechX Robotics · Tustin, CA

- Managed full P&L for a \$43,000+ revenue robotics education business across payroll, insurance, equipment, and tournament operations
- Designed competitive curriculum that led VEX Robotics teams to 1 World Championship and multiple State Championships

I.T. & Cybersecurity Support Intern

2023

Manhattan Beachwear · Cypress, CA

- Supported post-ransomware IT overhaul following a \$10M breach; deployed MFA, AI-endpoint protection, and secure remote access across all employee systems

PROJECTS

ClassMaker.ai — AI Scheduling Agent, Production-Deployed

April 2026

Built and shipped a full-stack AI scheduling agent at the Zero to Agent Global Hackathon — generates constraint-aware weekly class schedules within seconds using Groq's Llama 3.3, with CSV bulk import and plain-English chat refinement. Deployed live at classmaker.ai.

Titanic Survival Prediction — Dockerized ML API

April 2026

Containerized a machine learning pipeline as a REST API using FastAPI and Docker, achieving 81.56% test accuracy. Published the image to Docker Hub and documented the full MLOps deployment workflow.

LangChain Conversational AI — AWS Bedrock

March 2026

Built a stateful, context-aware chatbot using LangChain's ChatBedrock — implementing prompt templates, output parsers, conversation memory, and CSV document injection to power multi-turn AI conversations on Amazon Bedrock.

RAG System — Amazon Bedrock Knowledge Base

March 2026

Engineered a retrieval-augmented generation pipeline on Amazon Bedrock using Titan Embeddings for semantic search, OpenSearch Serverless as the vector store, and the RetrieveAndGenerate API to ground LLM responses in private documents.

House Price Prediction — Production REST API

March 2026

Deployed a Random Forest regression model as a production REST API using FastAPI and Pydantic schema validation, with startup model loading via joblib and a /health endpoint for monitoring.

Customer Churn Prediction — Beta Bank

January 2026

Built a binary classification model using feature engineering, SMOTE oversampling, and GridSearchCV across 108 hyperparameter combinations. Achieved $F1 = 0.6197$ on a 20% minority-class imbalanced dataset.

Big Data Benchmarking — Apache Spark on AWS EMR

April 2024

Benchmarked Apache Spark and Hadoop across EC2 instance types on Amazon EMR to identify the optimal distributed computing configuration. Delivered cost/performance analysis via CloudWatch metrics.

Serverless AI Application — AWS Lambda + Bedrock

February 2026

Connected Amazon Bedrock to a frontend via AWS Lambda and API Gateway to generate AI-powered flashcards from study notes — fully serverless, CORS-enabled, and prompt-engineered.

Megaline Prepaid Plan Statistical Analysis

March 2026

Analyzed 500 clients across 5 merged tables to compute per-user monthly revenue; confirmed via two-sample t-test that Surf generates significantly more revenue than Ultimate (avg \$50.33 vs \$47.31, $p \approx 0$).

Mobile Plan Recommendation Engine

December 2025

Random Forest classifier recommending the right Megaline mobile plan — 81.8% test accuracy, beating the 75% target by 6.8 percentage points.

Chicago Taxi Market Analysis

November 2025

SQL analysis and hypothesis testing proving bad weather significantly extends Loop-to-O'Hare taxi rides by 20.6% (t = 6.84, p ≈ 0).

Video Game Sales Analysis

July 2025

Mined 16,715 game records to build a data-backed 2017 advertising strategy — hypothesis-tested platform and genre preferences across NA, EU, and Japan markets.

Used Car Price Explorer — Interactive Web App

May 2025

Built and deployed an interactive Streamlit web app analyzing 51,500+ used vehicle listings — visualizing price distributions, condition trends, and odometer-to-price relationships. Live on Render.

IMDb "Golden Age" TV Analysis

April 2025

Tested whether highly-rated "Golden Age" TV shows also receive the most IMDb votes — cleaned and filtered messy real-world data before confirming the hypothesis.

Instacart Customer Behavior EDA

2024

Cleaned and analyzed 4.5M order records across 206K customers to uncover peak shopping windows, reorder rhythms, and top-ordered items across product departments.

VEX AI Robotics Competition Enhancement

February 2024

Proposed SpotFi localization, RGB-D 3D mapping, and P2P decentralized coordination to solve the limited field-of-view problem in autonomous VEX AI robots. (Georgia Tech CS6675)

AI Fairness in Housing Lending

Georgia Tech CS6603

Applied Disparate Impact and Statistical Parity Difference metrics to 2.5M Fannie Mae mortgage records across Race and Gender; measured whether Reweighting bias mitigation survives classifier training. (Georgia Tech CS6603)

Terraform Custom Provider — AWS S3 Static Sites

December 2023

Built a custom Terraform provider in Go with full CRUD support, deployed two live static sites via reusable S3 + CloudFront modules, and managed infrastructure state through Terraform Cloud.

CERTIFICATIONS

AWS Certified Cloud Practitioner

2026

Amazon Web Services

AWS Cloud Institute — Cloud Application Developer

Expected 2027

Amazon Web Services

EDUCATION

M.S. Computer Science — Specialization in AI

Expected 2028

Georgia Institute of Technology

AI and Machine Learning Bootcamp

Expected 2026

TripleTen

M.A. Teaching — Science Education

2011

University of Southern California